Data Mining Chapter 5 Resampling Techniques

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Exercise 1

Using basic statistical properties of the variance, as well as single-variable calculus, derive (5.6). In other words, prove that α given by (5.6) does indeed minimize

$$\operatorname{Var}\left(\alpha X + (1-\alpha)Y\right).$$

Answer

$$\begin{aligned}
\operatorname{Var}\left(\alpha X + (1-\alpha)Y\right) &= \alpha^2 \operatorname{Var}(X) + (1-\alpha)^2 \operatorname{Var}(Y) + 2\alpha(1-\alpha)\operatorname{Cov}(X,Y) \\
&= \alpha^2\sigma_X^2 + (1-\alpha)^2\sigma_Y^2 + 2\alpha(1-\alpha)\sigma_{XY} \\
&= \alpha^2\sigma_X^2 + (1-2\alpha+\alpha^2)\sigma_Y^2 + 2(\alpha-\alpha^2)\sigma_{XY}.
\end{aligned}$$

We now take the first derivative of $\operatorname{Var}(\alpha X + (1-\alpha)Y)$ relative to α and set it equal to 0 to get the point where the value of $\operatorname{Var}(\alpha X + (1-\alpha)Y)$ is minimum.

$$\begin{aligned}
\frac{\partial}{\partial\alpha}\operatorname{Var}\left(\alpha X + (1-\alpha)Y\right) &= 2\alpha\sigma_X^2 + (-2+2\alpha)\sigma_Y^2 + 2(1-2\alpha)\sigma_{XY} \\
&= 2\alpha\left(\sigma_X^2 + \sigma_Y^2 - 2\sigma_{XY}\right) - 2\left(\sigma_Y^2 - \sigma_{XY}\right) = 0.
\end{aligned}$$

Therefore,

$$\boxed{\alpha = \frac{\sigma_Y^2 - \sigma_{XY}}{\sigma_X^2 + \sigma_Y^2 - 2\sigma_{XY}}}$$

To check if this value is a minimum, we take the second derivative of $\operatorname{Var}(\alpha X + (1-\alpha)Y)$.

$$\begin{aligned}
\frac{\partial^2}{\partial\alpha^2}\operatorname{Var}\left(\alpha X + (1-\alpha)Y\right) &= 2\left(\sigma_X^2 + \sigma_Y^2 - 2\sigma_{XY}\right) \\
&= 2\operatorname{Var}(X-Y) \\
&\geq 0.
\end{aligned}$$

Since the second derivative is non-negative, the critical value of α minimizes $\operatorname{Var}(\alpha X + (1-\alpha)Y)$.

Exercise 9

We will now consider the Boston housing data set, from the ISLR library.

(a) Based on this data set, provide an estimate for the population mean of `medv`. Call this estimate $\hat{\mu}$.
Answer
The population mean is estimated using the sample mean of `medv`:

$$\hat{\mu} = \frac{1}{n}\sum_{i=1}^{n} \text{medv}_i.$$

Using the Boston housing data, the notebook obtains

$$\boxed{\hat{\mu} = 22.5328}$$

Thus, the estimated mean median value of owner-occupied homes is approximately 22.53.

(b) Provide an estimate of the standard error of $\hat{\mu}$. Interpret this result.

Hint: We can compute the standard error of the sample mean by dividing the sample standard deviation by the square root of the number of observations.
Answer
The notebook estimates the standard error of the sample mean using

$$SE(\hat{\mu}) = \frac{\text{sd(medv)}}{\sqrt{n}}.$$

The calculation in the notebook gives

$$\boxed{SE(\hat{\mu}) = 0.40846}.$$

This means that the estimated sample mean of `medv` would typically vary by approximately 0.41 units from sample to sample due to sampling variation.

Note: The notebook uses NumPy's default standard deviation calculation (`np.std`), which uses $ddof = 0$.

(c) Now estimate the standard error of $\hat{\mu}$ using the bootstrap. How does this compare to your answer from (b)?
Answer
A bootstrap procedure was used to estimate the standard error of $\hat{\mu}$. The procedure generated 10,000 bootstrap samples, each obtained by sampling the observed `medv` values with replacement.

The mean of the bootstrap estimates was

$$22.5286,$$

which is very close to the original sample mean of 22.5328.

The standard deviation of the 10,000 bootstrap estimates was

$$\boxed{SE_{\text{bootstrap}}(\hat{\mu}) = 0.40738}.$$

The two estimates are therefore very similar:

$$SE_{\text{formula}} = 0.40846, \qquad SE_{\text{bootstrap}} = 0.40738.$$

The difference is very small, indicating that the bootstrap provides a result consistent with the direct standard-error calculation.

(d) Based on your bootstrap estimate from (c), provide a 95% confidence interval for the mean of `medv`. Compare it to the results obtained by using `Boston['medv'].std()` and the two standard error rule (3.9).

Hint: You approximate a 95% confidence interval using the formula

$$[\hat{\mu} - 2SE(\hat{\mu}), \hat{\mu} + 2SE(\hat{\mu})] .$$

Answer
Using the bootstrap estimate of the standard error and the approximate two-standard-error rule,

$$[\hat{\mu} - 2SE(\hat{\mu}), \hat{\mu} + 2SE(\hat{\mu})] ,$$

we obtain

$$\begin{aligned}
\text{Lower bound} &= 22.5328 - 2(0.40738) \\
&\approx 21.718, \\
\text{Upper bound} &= 22.5328 + 2(0.40738) \\
&\approx 23.348.
\end{aligned}$$

Therefore, the approximate 95% confidence interval is

$$\boxed{[21.718,\ 23.348]}.$$

This interval suggests that the population mean of `medv` is reasonably expected to lie between approximately 21.72 and 23.35.

The interval is based on the bootstrap standard error and the approximate two-standard-error rule requested in the exercise.

(e) Based on this data set, provide an estimate for $\hat{\mu}_{\text{med}}$, for the median value of `medv` in the population.

Answer

The population median is estimated by the sample median of `medv`. The notebook gives

$$\boxed{\hat{\mu}_{\text{med}} = 21.2}.$$

Thus, the estimated median value of `medv` is 21.2.

(f) We now would like to estimate the standard error of $\hat{\mu}_{\text{med}}$. Unfortunately, there is no simple formula for computing the standard error of the median. Instead, estimate the standard error of the median using the bootstrap. Comment on your findings.

Answer

There is no simple standard formula analogous to the standard error of the sample mean that is used here for the median. Therefore, the notebook estimates the standard error using the bootstrap.

Using 10,000 bootstrap samples, the mean of the bootstrap median estimates was

$$21.1873,$$

which is close to the original sample median of 21.2.

The standard deviation of the bootstrap median estimates was

$$\boxed{SE_{\text{bootstrap}}(\hat{\mu}_{\text{med}}) = 0.37980}.$$

Therefore, the bootstrap indicates that the estimated median has a standard error of approximately 0.38.

For completeness, the notebook also calculated an approximate two-standard-error interval:

$$[21.2 - 2(0.37980), 21.2 + 2(0.37980)]\,,$$

giving

$$\boxed{[20.44,\ 21.96]}.$$

The bootstrap is useful here because it provides a practical way to assess the sampling variability of the median without relying on a simple closed-form standard-error formula.

(g) Based on this data set, provide an estimate for the tenth percentile of `medv` in Boston census tracts. Call this quantity $\hat{\mu}_{0.1}$.
Answer
The tenth percentile of `medv` was calculated using the `np.percentile()` function.

The notebook gives

$$\boxed{\hat{\mu}_{0.1} = 12.75}.$$

Thus, approximately 10% of the observed Boston census tracts have a `medv` value at or below 12.75.

(h) Use the bootstrap to estimate the standard error of $\hat{\mu}_{0.1}$. Comment on your findings.
Answer

The tenth percentile was estimated for each of 10,000 bootstrap samples. The mean of the bootstrap tenth-percentile estimates was

$$12.7521,$$

which is very close to the original estimate of 12.75.

The standard deviation of the bootstrap estimates was

$$\boxed{SE_{\text{bootstrap}}(\hat{\mu}_{0.1}) = 0.49685}.$$

Therefore, the estimated standard error of the tenth percentile is approximately 0.50.

The bootstrap estimate shows that the tenth percentile has somewhat greater sampling variability than the estimated mean and median. This is reasonable because percentile estimates, particularly those away from the center of the distribution, can be more sensitive to the particular observations included in a sample.